Lab Assignment: Introduction to Object-Oriented PHP

Web Information Systems

Lab Topic

Practicing Classes, Objects, Constructors, Access Modifiers, and Inheritance

Difficulty

Beginner

Estimated Time

60–90 minutes

1. Lab Objectives

After completing this lab, students should be able to:

- Create a simple PHP class.
- Create an object from a class.
- Create and call methods.
- Create properties inside a class.
- Use a constructor to initialize properties.
- Use $this to access object properties.
- Understand public , private , and protected . •
  Create a simple parent and child class using inheritance.

Part A — Simple Class and Object

Create a PHP file named:

```
lab_oop.php
```

Create a class called:

Student

The class should contain one method called:

sayHello()

The method should display:

Hello! I am a student.

Requirements

1. Create the Student class.
2. Create the sayHello() method.
3. Create an object called $student1 .
4. Call the sayHello() method.

Starter Code

```php
<?php

class Student
{
	function sayHello()
	{
		// Write your code here
	}
}


// Create an object here


// Call the method here


?>
```

Expected Output

```
Hello! I am a student.
```


Part B — Class with Constructor

Now improve the Student class.

The student should have three properties:

```
name
studentId
department
```

Use a constructor to give values to these properties.

Requirements

Create these properties:

```
public $name;
public $studentId;
public $department;
```

Create a constructor that receives:

```
$name
$studentId
$department
```

Store the values using $this .

Create a method called:

```
showInfo()
```

The method should display all student information.

Starter Code

```php
<?php
```



```php
class Student
{
	public $name;
	public $studentId;
	public $department;

	function __construct($name, $studentId, $department) {
		// Store the values in the properties
	}

	function showInfo()
	{
		// Display student information
	}
}

$student1 = new Student(
	"Ahmad",
	1001,
	"Computer Science"
);

$student1->showInfo();

?>
```

Expected Output

```
Name: Ahmad
Student ID: 1001
Department: Computer Science
```

Part C — Create Another Object

Using the same Student class, create another student. Use the following information:

Name: Sara
Student ID: 1002
Department: Information Systems

Create:

```
$student2
```

Then call:

```
$student2->showInfo();
```

Expected Output

Name: Sara
Student ID: 1002
Department: Information Systems

Question

How many classes did you create?

Answer: ______________________

How many objects did you create?

Answer: ______________________

```
{
```

Part D — Access Modifiers

Create a new class called:

```
BankAccount
```

The class should contain:


```
public $ownerName;

private $balance;
```

Use a constructor to initialize both properties.

Requirements

Create:

```
public $ownerName;
private $balance;
```

Create a constructor.

Create a method:

```
showBalance()
```

The method should display the balance.

Starter Code

```
<?php

class BankAccount
{
        public $ownerName;
```

```
        private $balance;

        function __construct($ownerName, $balance) {
                // Store the values
        }

        function showBalance()
        {
                // Display the balance
        }
```



```
}

$account1 = new BankAccount(
        "Ahmad",
        5000
);

echo "Owner: " . $account1->ownerName . "<br>";

$account1->showBalance();

?>
```

Expected Output

```
Owner: Ahmad
Balance: 5000
```

Try This

After running the program successfully, try:

```
echo $account1->balance;
```

Question

Does it work?

Yes / No

Why?

____

____

Hint

Remember:

public → can be accessed outside the class private → can only be

accessed inside the same class

Part E — Simple Inheritance

Create a parent class called:

Person

The Person class should have:

public $name;

Create a constructor to receive the name.

Create a method:

introduce()

```
}
```

Create another class called:

```
Student
```

The Student class should inherit from Person .

Use:

```
extends
```

The Student class should have another method:


```
study()
```

Starter Code

```php
<?php

class Person
{
        public $name;

        function __construct($name)
        {
                // Store the name
        }

        function introduce()
        {
                // Display the person's name
        }
}

class Student extends Person
{
        function study()
```

```
    {
        // Display that the student is studying }
}


$student1 = new Student("Ahmad");


$student1->introduce();

$student1->study();

?>
```


Expected Output

```
My name is Ahmad
Ahmad is studying.
```

Part F — Understanding Inheritance

Answer the following questions.

Question 1

Which class is the **parent class**?

____

Question 2

Which class is the **child class**?

____

Question 3

Which keyword creates the inheritance relationship?

____

Question 4

The introduce() method was written inside which class?

____

Question 5

Can the $student1 object call introduce() ?

Yes / No

Question 6

Why can $student1 use introduce() even though it is not written inside the Student class?

____

____

Part G — Small Independent Exercise

Create your own simple OOP program using **Vehicle and Car**.

Parent Class: Vehicle

Create:

```
class Vehicle
```

It should contain:

```
protected $brand;
```

Create a constructor that receives the brand.

Create a method:

```
start()
```

The method should display:

```
The vehicle is starting.
```


Child Class: Car

Create:

```
class Car extends Vehicle
```

Create a method:

```
showBrand()
```

It should display the car brand.

Create an Object

Create:

```
$car1 = new Car("Toyota");
```

Call:

```
$car1->start();

$car1->showBrand();
```

Expected Output

```
The vehicle is starting.
Car brand: Toyota
```

Lab Submission Requirements

Submit **one PHP file** containing your completed exercises. File name:


```
StudentID_OOP_Lab.php
```

Example:

```
2025001_OOP_Lab.php
```

Your program should:

- Run without PHP errors.
- Use classes and objects correctly.
- Include at least one constructor.
- Use $this correctly.
- Include public and private properties.
- Include one inheritance example.
- Use meaningful class, property, and method names.

Suggested Marking Scheme

Task	Marks
Part A — Class, object, and method	2

Part B — Constructor and properties	2
Part C — Multiple objects	1
Part D — Access modifiers	2
Part E — Inheritance	2
Code correctness and organization	1
Total	**10**

Quick Reminder

Class
↓
Object
↓
Method
↓

Properties
↓
Constructor
↓
$this
↓
Access Modifiers
↓
Inheritance

For this first lab, focus on understanding **how the pieces connect together** rather than writing a large program.